

Time: 25 Mins.

Student Id:

Obtained Marks:

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- Q1.** Is it possible to perfectly reconstruct the original analog signal after converting it into digital form? Explain your answer with appropriate justification, considering the concepts of sampling and quantization. **03**
- Q2.** Suppose, it is given $V_{in} = 0.4 \text{ Volt}$, $V_{ref} = 1 \text{ Volt}$ and **8-bit** of resolution for a *Successive Approximation A/D* converter. Find the **8-bit** digital output for the given V_{in} . Also find the analog value for that digital output using the *Weighted Sum D/A* conversion method. **05+02**
- Q3.** A 5-bit D/A converter produces $V_{out} = 0.2 \text{ V}$ for a digital input of 00001. Find the value of V_{out} for an input of 11111. **05**
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